

[← Go Back](#)

Hardware

[Setting Up the Opta™ for Data Logging on Opta™](#)

[Memory Partitioning for Use with the Arduino IDE](#)

[Modbus TCP On Opta™ Using PLC IDE](#)

[Tank Thermoregulation with Portenta Machine Control & Opta™](#)

[Energy Management with Opta™](#)

[Tank Level Monitoring with the Opta™](#)

[Home Automation with Opta™](#)

[Retrieve the Opta™ MAC Address](#)

[Opta™ Digital Expansions Usage with the PLC IDE](#)

[Conveyor Belt Sorting with Remote Monitoring Using Opta™](#)

[Air Compressor Remote Condition Monitoring with Opta™](#)

[Getting Started with Modbus TCP on Opta™](#)

[Opta™ Analog Expansion Usage with the PLC IDE](#)

Modbus RTU On Opta™ Using PLC IDE

[Packaging and Labeling with Remote Monitoring Using Opta™](#)

[Edge AI Motor Anomaly Detection with Opta™ & Nicla Sense ME](#)

[Integrate Arduino Opta with Ignition HMI Through OPC-UA](#)

Home / Hardware / Opta / **Modbus RTU On Opta™ Using PLC IDE**

Modbus RTU On Opta™ Using PLC IDE

Learn how to set and enable Modbus RTU on Opta™ using Arduino® PLC IDE.

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Overview

Opta™ has scalable industrial-grade hardware with a wide range of connectivity choices. Opta™ is enhanced by the Arduino PLC IDE software, which makes the most of the device for solid field deployments. Opta™ supports Modbus protocols, and the Arduino PLC IDE makes it simple to implement them.



System overview of Opta™ with Modbus RTU using PLC IDE

In this tutorial, you will learn how to implement Modbus RTU-based communication between two Opta™ devices using Arduino PLC IDE.

ON THIS PAGE

Overview

Goals	
Required Hardware and Software	—
Hardware Requirements	
Software Requirements	
Modbus Protocol	
Modbus RTU & PLC IDE	
Instructions	—
Setting Up the Arduino PLC IDE	
Hardware Setup	
Workspace Pre-Configuration	+
Project Overview	+
Testing Modbus RTU Communication Between Opta™ Devices (PLC IDE)	
Conclusion	+

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Learn how to configure the workspace environment to work with Modbus RTU using Arduino PLC IDE.

Learn how to configure Modbus RTU for Opta™ using Arduino PLC IDE.

Learn to verify that Opta™ has been correctly set up using a Modbus RTU communication example.

Required Hardware and Software

Hardware Requirements

[Opta™](#) (x2)

[USB-C® cable](#) (x2)

Cable with either specification for RS-485 connection (x3):

STP/UTP 24-18AWG
(Unterminated) 100 - 130
Ω rated

STP/UTP 22-16AWG
(Terminated) 100 - 130 Ω
rated

Software Requirements

[Arduino PLC IDE Installer](#)

If you have an Opta™, you do not need any license key to activate your product. Learn more about this in [License Activation With Pre-Licensed Products \(Opta™\)](#).

[Opta™ Modbus RTU PLC IDE Project Example File](#)

Modbus Protocol

Modbus is an open, client/server architecture-based serial communication protocol free of licensing restrictions. It is commonly employed in industrial electrical equipment, particularly in Building Management Systems (BMS) and Industrial Automation Systems (IAS).

It was released by Modicon (now Schneider Electric) in 1979. Since then, it has become the de facto industry standard for industrial electronic devices communicating with PLCs.

In Supervisory Control and Data Acquisition (SCADA) systems, a Remote Terminal Unit (RTU) and a supervisory device are frequently connected via the Modbus communication protocol. Modbus messages have a straightforward 16-bit format and a Cyclic-Redundant Checksum (CRC) when communicating with electronic equipment.

For more information regarding the Modbus RTU protocol



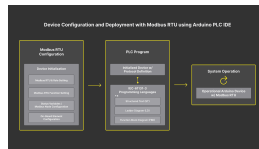
implementation on an Opta™, you may want to check out [the "Getting Started with Modbus RTU on Opta™" tutorial](#).

Modbus RTU & PLC IDE

This tutorial will guide you through setting up two Opta™

the Arduino PLC IDE and help us understand the overall implementation process.

The following diagram illustrates how Opta™ is configured and deployed with Modbus RTU:



Modbus RTU
Implementation with
Arduino PLC IDE

The entire procedure is divided into three distinct stages:

Modbus RTU

Configuration is the foundational step in initializing the Opta™ device with Modbus RTU and other intrinsic properties.

The Modbus RTU role, Client or Server, is assigned to the Opta™ device during this stage. It includes the 'Baud rate' and 'Serial Mode' configurations, which are essential for Modbus RTU communication.

Based on the device setting within the Modbus RTU, either the 'Status variables' are delineated, or the Modbus node is defined to determine the communicating devices using this protocol.

PLC Program, established after device setup, is based on Modbus RTU and additional features. A

elimination of intricate configurations or specific programming tied to Modbus RTU within the PLC code.

The procedure is designed for ease of use. By simply using predefined variables in the PLC code, the system handles data transfer through Modbus RTU. This approach benefits from the device's prior setup, minimizing repetitive tasks and promoting effective communication.

System Operation

represents the anticipated outcome post the Modbus RTU configuration and the execution of the PLC program based on the developer's designed logic. Consequently, we can observe the device communicating with other devices through Modbus RTU.

The diagram presented shows the uniformity of the Modbus RTU setup across devices like Opta™ and the Portenta Machine Control using the PLC IDE. One main advantage of this system is its adaptability. Regardless of the specifics of the Modbus RTU setup, the PLC program consistently performs effectively. Its design ensures that it can be used in many different environments.

In addition, when developing

The system adheres to the IEC61131-3 standard, allowing one to select from the languages outlined in this standard. It ensures a balance between user convenience and accurate code development.

We can now delve into the specifics after providing an overview of the entire process.

Instructions

Setting Up the Arduino PLC IDE

To get the Arduino PLC IDE software, go to the [official software website of the Arduino PLC IDE](#) and choose to download the PLC IDE installer file. The software is named

Arduino PLC IDE Installer.



Arduino PLC IDE Software Download Section

The software requires **Windows 10** or a newer operating system version for the x64 architecture.

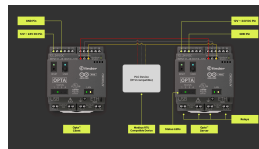
The Arduino PLC IDE installer contains the IDE and all the required drivers, libraries, and cores. The following sections will help you install the software properly.



For more details regarding Arduino PLC IDE setup, please take a look at [Arduino PLC IDE Setup and Board's License Activation](#) tutorial.

Hardware Setup

The two Opta™ devices will communicate using Modbus RTU, which is enabled by using the RS-485 interface for both devices. The following image shows the connection diagram for working with two Opta™ devices.



RS-485 interface
between Opta™
devices for Modbus
RTU

The Modbus RTU communication network can be scaled up by integrating additional protocol-compatible devices such as Opta™ or Portenta Machine Control.

Opta™ does not come equipped with integrated termination resistors; therefore, it is essential to follow the Modbus protocol specifications, which require the addition of termination resistors.

 Should you experience inconsistencies in data transmission when using Opta™ alongside other Modbus RTU-compatible devices, it is advised to switch the A(-) and B(+) lines and initiate the process again.

Workspace Pre-Configuration

Some considerations must be made before properly enabling and using Modbus RTU on Opta™ using PLC IDE for appropriate Modbus RTU operation. The following subsections will explain such aspects.

 Check out [this tutorial](#) to familiarize yourself with the Arduino PLC IDE environment.

Opta™ Basic Configuration

The Modbus RTU communication for Opta™ does not require special pre-configuration. You will only have to make a manual sketch download with the desired protocol role, its properties, and its onboard elements to be used. These onboard elements can be status LEDs and relays.

configuration changes are required.

Modbus RTU Client and Server Mode

The PLC IDE software provides the option to set Opta™ as either a Modbus RTU Client, Server, or none of the previous modes. The option is configurable under `RS485 SerialPort` listed within the `Resources` tab.

Opta™ set as a Modbus RTU Client will provide 'Baud Rate' and 'Serial Mode' settings. The baud rate is available from 600 b/s to 115200 b/s. The serial mode offers a set of options conformed with the following elements:

Parity: No Parity, Even Parity, Odd Parity

Data Bits: 8 Data Bits
(Only option for Data Bits)

Stop Bits: 1 - 2 Stop Bits

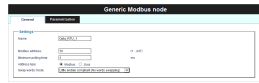
On the other hand, Opta™ set as a Modbus RTU Server requires an additional configuration called 'Slave Settings'. It will ask you to define the Modbus address with a range between `1 .. 247`.

If you disable Modbus RTU for Opta™, you can select the option `none`. This option hides every Modbus RTU configuration parameter, turning off the protocol mode for Opta™.

General Modbus Node Configuration

The General Modbus Node

information regarding the devices compatible with the Modbus messaging service.



Arduino PLC IDE - General Modbus Node Configuration

It will require you to fill in basic information under the **General** tab and parameters to manage under the **Parametrization** tab. The basic information consists of:

- Name of the device to be communicated under Modbus protocol

- Modbus address (1..247)

- Minimum polling time [ms]

- Address type

- Swap words mode

This information will help identify and correctly communicate with the target device.

PLC IDE Modbus Custom Editor

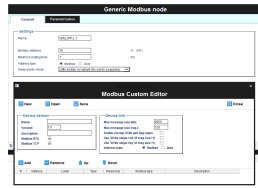
The Modbus Custom Editor allows you to define a Modbus node with predefined parameters and variables. You can later add it using the **Add** option under

RS485 SerialPort.

To open the Modbus Custom Editor window, go to

Tools -> Run Modbus Custom Editor

on PLC IDE.



Arduino PLC IDE - Modbus Custom Editor Configuration

It is helpful to have frequently deployed device configurations stored that are compatible with the Modbus protocol.

Project Overview

The example project will test the Modbus RTU connection between two Opta™ devices after you have prepared the necessary prerequisites and tools.

The example project will slightly modify its default example code to incorporate a counter (`cnt`) variable, creating a live handshake verification procedure between two Opta™ devices and broadcasting the counter data.

Based on the counter data it receives from the "Modbus RTU Server Opta™," the "Modbus RTU Client Opta™" will activate the relay and the status LED. Using the role-specific sections, you may learn how to configure the Modbus RTU role for each Opta™ device.

You may access the entire example project [here](#) to test it immediately. Every setting and component is ready to be assembled and uploaded to the corresponding Opta™.

Opta™ according to its function in a Modbus RTU connection.

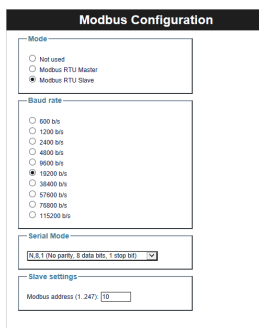
Modbus RTU Server Opta™

To set Opta™ as a Modbus RTU Server, navigate to the `RS485 SerialPort` tab in the PLC IDE's `Resources` panel. A window named `Modbus Configuration` will open, and we must select the `Modbus RTU Slave` option. For the tutorial example, we will use the following properties for server Opta™:

Baud Rate: 19200 b/s

Serial Mode: N,8,1 (No parity, 8 data bits, 1 stop bit)

Slave settings (Modbus address): 10



Arduino PLC IDE -
Opta™ Modbus RTU
Server Configuration

Alternative values can be used per requirements if needed.

The following image shows the `Status variables (volatile)` window. Here, we will define the `cnt` variable and assign its access address and datatype for Modbus RTU transmission.



Arduino PLC IDE -

The `cnt` status variable uses the following parameters:

Address: 25000 (dec) /
0x61A8 (hex)

Name: cnt

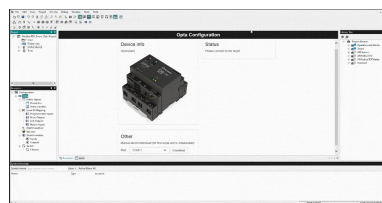
PLC type: INT

With these settings ready, you **must go** to

`Resources -> Opta`, select the corresponding port, and begin the `Manual sketch download` process. Then you **must go** to

`On-line -> Set up Communication`

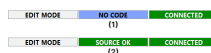
and activate Modbus RTU with the higher USB port number assigned for Opta™.



Proceed with

`On-line -> Connect`, and it

will establish communication between your computer and the Opta™ server. If everything is fine, you can observe the message at the lower right corner of the PLC IDE software stating that it is connected.



Arduino PLC IDE -
Device Connection
Status

Symbols `(1)` and `(2)` denote the connection statuses: Opta™ connected without a PLC code and Opta™ connected with an available PLC code, respectively.

Next, the main PLC code, also

to Opta™. To do this, navigate to the `main` tab within the `Project` panel and input the following code:

```
1 cnt := cnt + 1;  
2  
3 IF cnt >= 2750 THEN  
4   cnt := 0;  
5 END_IF;
```

The Opta™ server device's task runs a simple counter and resets whenever the counter reaches `2750`. Use the `Download PLC code` option or press `F7` to initiate the code's compilation and uploading process to Opta™. A successful upload will resemble the image provided below.



Arduino PLC IDE -
Opta™ Server Main
Code

Upon completing these steps, you will have successfully configured an Opta™ device as a Modbus RTU Server. The following section will guide you through setting up another Opta™ as a Modbus RTU Client.

Modbus RTU Client Opta™

To set Opta™ as a Modbus RTU Client, navigate to the `RS485 SerialPort` tab in the PLC IDE's `Resources` panel. The `Modbus Configuration` panel will open, and we must select the `Modbus RTU Master` option. For the tutorial example.

Baud Rate: 19200 b/s

Serial Mode: N,8,1 (No parity, 8 data bits, 1 stop bit)

Modbus Configuration

Mode

☐ Not used

☒ Modbus RTU Master

☐ Modbus RTU Slave

Baud rate

☐ 600 b/s

☐ 1200 b/s

☐ 2400 b/s

☐ 4800 b/s

☐ 9600 b/s

☒ 19200 b/s

☐ 38400 b/s

☐ 57600 b/s

☐ 76800 b/s

☐ 115200 b/s

Serial Mode

N,8,1 (No parity, 8 data bits, 1 stop bit)

Arduino PLC IDE -
Opta™ Modbus RTU
Client Configuration

Alternative values can be used per requirements if needed.

To establish communication with a configured Modbus RTU Server on an Opta™ device, add a Modbus node by right-clicking the **RS485 SerialPort** tab in the **Resources** panel. After right-clicking, select the 'Add' option that appears. This action will allow you to introduce a 'Generic Modbus' node into the configuration. For this example, configure the node with the following parameters:

Name: Opta_RTU_1

Modbus address: 10

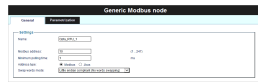
Minimum polling time: 1 ms

Address type: Modbus

Swap words mode: Little-endian compliant (No words swapping)

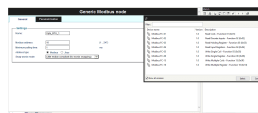
For these settings, follow the configuration pattern used for the Modbus RTU Server Opta™. The most essential detail to set is the Modbus address. Figure

given to the server Opta™ or any other compatible device should you add more nodes. The setup should resemble the image provided:



Arduino PLC IDE -
Opta™ Client Node

Once you have established the Modbus node for the client Opta™, it is time to determine the Modbus function to fetch the counter (cnt) data from the server Opta™. Right-click on `Opta_RTU_1` or any other name you set with, and the 'Add' option will show up, displaying a device catalog window with all available Modbus functions:



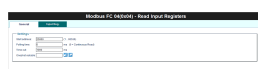
Arduino PLC IDE -
Modbus Functions

To retrieve counter information from the server Opta™, select the 'Modbus FC-04 (Read Input Registers)' function. The 'General' tab needs to be configured with the following parameters to ensure correct data access:

Start address: 25000

Polling time: 0 ms
(Continuous Read)

Timeout: 1000 ms



Arduino PLC IDE -
Opta™ Client Modbus

Subsequently, you must define a variable to store the counter data retrieved from the server Opta™. To do this, navigate to the Modbus function configuration interface's **Input Reg.** tab. Introduce a variable named **counter_rec** to capture the data transmitted through the protocol.

The following image shows a visual representation of the anticipated configuration:

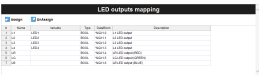


Arduino PLC IDE -
Opta™ Client Modbus
Function of the Node
(Input Reg.)

In this tutorial's demonstration, the client Opta™ is configured to use status LEDs and relays to respond to particular actions. We have the flexibility to assign variables to these status LEDs. For the present example, we will use designations ranging from **LED1** to **LED4** for the corresponding status LEDs.

For example, to use the **relay_1** and **LED1** variables for the first relay and status LED, respectively, these variables should be defined under the 'Local IO Mapping' section, specifically within 'Relay Outputs' and 'LED Outputs'. Neglecting this step will cause the program compilation to flag errors because of these undefined designations.

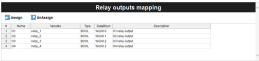
The following image shows a glimpse into how the configuration should appear



LED	Modbus Address	Variable Name
LED 1	0x0001	LED1_ON
LED 2	0x0002	LED2_ON
LED 3	0x0003	LED3_ON
LED 4	0x0004	LED4_ON
LED 5	0x0005	LED5_ON
LED 6	0x0006	LED6_ON
LED 7	0x0007	LED7_ON
LED 8	0x0008	LED8_ON
LED 9	0x0009	LED9_ON
LED 10	0x000A	LED10_ON
LED 11	0x000B	LED11_ON
LED 12	0x000C	LED12_ON
LED 13	0x000D	LED13_ON
LED 14	0x000E	LED14_ON
LED 15	0x000F	LED15_ON
LED 16	0x0010	LED16_ON

Arduino PLC IDE -
Opta™ Client Status
LED Table

Relays also need specific designations to reference them in the main PLC code. Below is a table presenting the variable names designated for the relays:



Relay	Modbus Address	Variable Name
Relay 1	0x0011	RELAY1_ON
Relay 2	0x0012	RELAY2_ON
Relay 3	0x0013	RELAY3_ON
Relay 4	0x0014	RELAY4_ON
Relay 5	0x0015	RELAY5_ON
Relay 6	0x0016	RELAY6_ON
Relay 7	0x0017	RELAY7_ON
Relay 8	0x0018	RELAY8_ON
Relay 9	0x0019	RELAY9_ON
Relay 10	0x001A	RELAY10_ON
Relay 11	0x001B	RELAY11_ON
Relay 12	0x001C	RELAY12_ON
Relay 13	0x001D	RELAY13_ON
Relay 14	0x001E	RELAY14_ON
Relay 15	0x001F	RELAY15_ON
Relay 16	0x0020	RELAY16_ON

Arduino PLC IDE -
Opta™ Client Relay
Table

The program outlined below is designed to retrieve counter data, oversee status LEDs, and handle the corresponding relays. The proper functioning of Modbus RTU communication ensures the efficient execution of the tasks, as mentioned earlier.

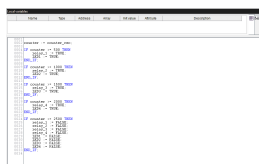
```

1  counter := counter
2
3  IF counter >= 500
4      relay_1 := TRU
5      LED1 := TRUE;
6  END_IF;
7
8  IF counter >= 1000
9      relay_2 := TRU
10     LED2 := TRUE;
11 END_IF;
12
13 IF counter >= 1500
14     relay_3 := TRU
15     LED3 := TRUE;
16 END_IF;
17
18 IF counter >= 2000
19     relay_4 := TRU
20     LED4 := TRUE;
21 END_IF;
22
23 IF counter >= 2500
24     relay_1 := FAL
25     relay_2 := FAL
26     relay_3 := FAL
27     relay_4 := FAL
28     LED1 := FALSE;

```

The variable `counter` serves as a global reference for the client Opta™. At the same time, `counter_rec` is the Modbus-specific variable that stores data fetched from the server Opta™. This variable was defined when configuring the `Read Input Registers` Modbus function.

Upon successful compilation and downloading of the main PLC code, the complete interface for the client Opta™ should resemble the image provided below:



Code

Finally, Opta™ is now ready as a Modbus RTU Client.

Testing Modbus RTU Communication Between Opta™ Devices (PLC IDE)

You can access the complete example project [here](#). You can download the compressed file, extract it, and use the pre-configured example project for your Opta™ devices.

Set both Opta™ devices running with the corresponding main PLC code with the hardware setup explained in [this section](#). You will be able to observe the following results on client Opta™ repeatedly:

Counter value = **500**: The status LED #1 and relay #1 will turn on

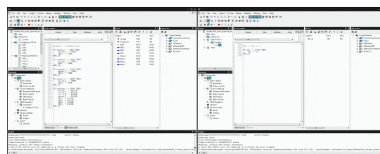
Counter value = **1000**: The status LED #2 and relay #2 will turn on

Counter value = **1500**: The status LED #3 and relay #3 will turn on

Counter value = **2000**: The status LED #4 and relay #4 will turn on

Counter value = **2500**: All status LEDs and relays will turn off

The following short clip briefly shows the expected behavior of the example project.



represents the server Opta™.

Conclusion

In this tutorial, you have learned to configure the workspace environment to work with Modbus RTU using Arduino PLC IDE and verified that Opta™ has been correctly set up. Modbus RTU communication is effective using an example project that controls the Opta™ device's onboard features, such as relays and status LEDs, based on a customized example.

Next Steps

Now that you have learned to implement the Modbus RTU between Opta™ devices using Arduino PLC IDE try adding additional Modbus RTU-compatible devices and creating a Modbus RTU communication network.

Further, explore the possibilities by combining the Opta™ device's onboard features with the Modbus RTU communication network and deploy it as an enhancement solution for industrial management systems.

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